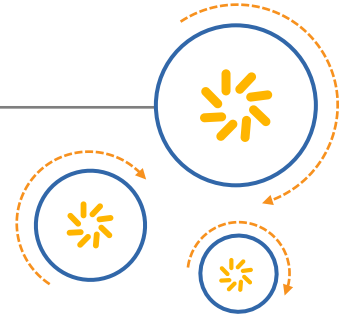




Qualcomm Technologies, Inc.



DragonBoard™ 410c based on Qualcomm® Snapdragon™ 410E processor

Enable USB Camera Application Note

September 2016

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Questions or comments: <https://www.96boards.org/DragonBoard410c/forum>

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LM80-P0436-22 Rev B

Revision history

Revision	Date	Description
B	September 2016	Update to 'E' part
A	July 2, 2015	Initial release.

Contents

- 1 Introduction 4**
 - 1.1 Purpose4
 - 1.2 Acronyms and abbreviations.....4
 - 1.3 USB camera used.....4
- 2 Testing USB Camera on Android Lollipop 5**
 - 2.1 Testing USB camera.....5
- EXHIBIT 1..... 7**

1 Introduction

1.1 Purpose

This application note explains how to enable the USB camera on the DragonBoard™ 410c development board.

1.2 Acronyms and abbreviations

Table 1 Acronyms and abbreviations

Acronym	Definition
APK	Application Package File
GND	Ground
HD	High Definition
SD	Secure Digital
TTL	Transistor Transistor Logic
UART	Universal Asynchronous Receiver Transmitter
USB	Universal Serial Bus

1.3 USB camera used

The Creative Cam Sync HD Webcam VF0770 was used for testing.



2 Testing USB Camera on Android Lollipop

2.1 Testing USB camera

The Android build used for testing is dated 6/18/2015 and available on <http://www.96boards.org/product/dragonboard410c/>

Android 5.1 (Lollipop)

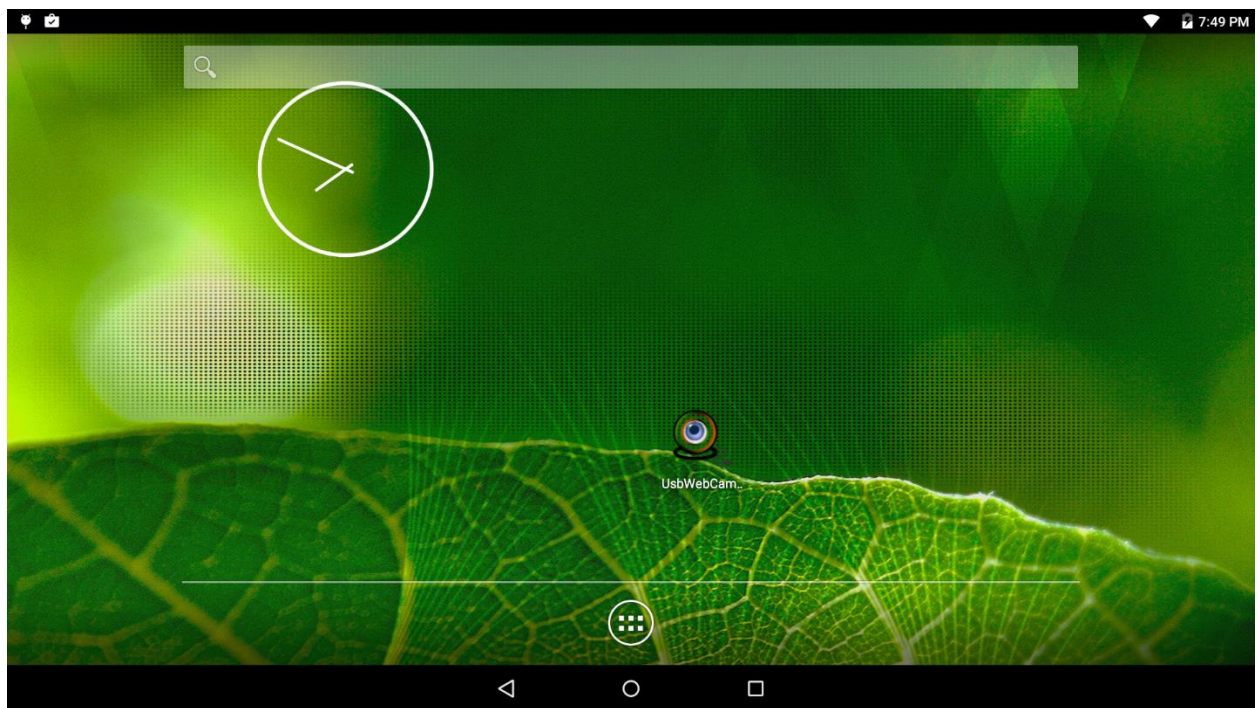
Linux Kernel 3.10

Release: 15.06

This build has the support for the USB camera enabled by default.

Download the SD card installation package and install the Android images. This should also work with the default Android images shipped with DragonBoard 410c.

The application used to verify USB camera is UsbWebCam APK.



You can use the “dmesg” on the serial console to detect the new USB camera device as soon as you connect it to one of the USB2.0 port on DragonBoard 410c. A new /dev/videoX device node should also be created. To see the console trace messages you need to have the USB to UART TTL 1.8V cable connected to J8 to pin 1 (GND), 11 (UART1_TX), 13 (UART1_RX). Set the baud rate to 115200. You cannot use the adb shell, as only device mode or host mode is supported at any time on DragonBoard 410c.

As soon as you click on the UsbWebCam icon you should be able to see the preview from the USB camera. You can also take snapshots or capture video.

NOTE: This application is for basic validation purposes only, there may be limitations with the functionality, in which case you would need to implement you own video dev application to stream and capture from the USB camera.

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